DO NOW — RECALL

Bubble sort: trace and implement

Name: _____ Class: _____ Date: _____

1. For CountDigits(582), list the calls and give the returned digit count.

2. How many times does a loop with i = 0; i < 4; i++ run?

3. For A=true and B=false, what is A && B?

4. After swapping neighbouring values 7 and 3 in [7,3,5], what is the array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Calls 582 → 58 → 5; returns 1 → 2 → 3. The digit count is 3.

2. Model answer

4 times: i is 0,1,2,3.

3. Model answer

false. Both must be true for AND.

4. Model answer

[3,7,5].

Vocabulary — know these terms

pass — One traversal of the remaining unsorted section.

adjacent — At neighbouring indices.

swap — Exchange two stored values.

comparison — Test used to determine whether a swap is required.

early exit — Stop when a pass makes no swaps.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Bubble sort always needs to make at least one swap.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why can the early-exit optimisation finish immediately on [1,2,3,4]?

NEW LEARNING

Read, annotate and explain

Compare neighbours

Compare positions j and j+1. Swap only if the earlier value is greater.

Finish one pass

The greatest unsorted value moves to the right-hand end of the unsorted section.

Repeat less work

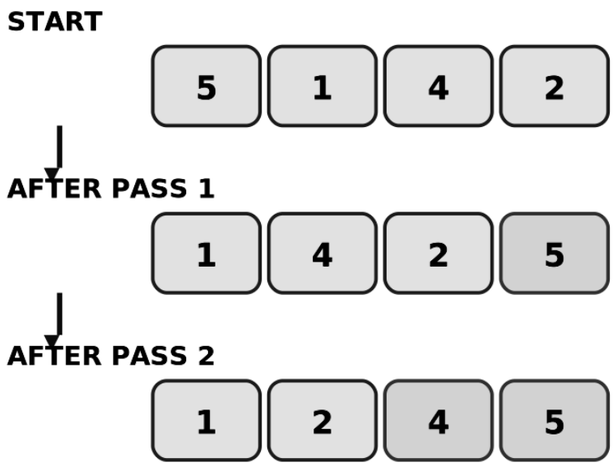
Each completed pass fixes one final position; an entire pass with no swaps proves the array is sorted.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: START 5,1,4,2; AFTER PASS 1 1,4,2,5; AFTER PASS 2 1,2,4,5

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Show one left-to-right pass on [7,3,6,2].

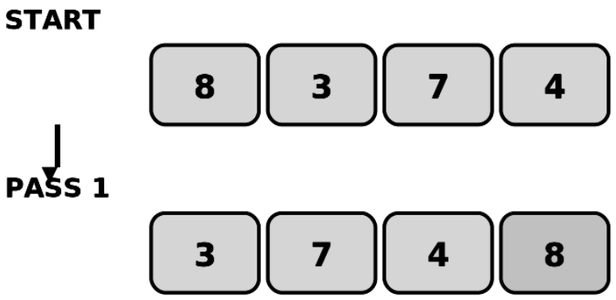
Question 2

Show two complete passes on [7,3,6,2].

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Complete one pass on [8,3,7,4].



Worked algorithm states: START 8,3,7,4; PASS 1 3,7,4,8

I DO — WORKED EXAMPLE

Follow the worked method

Trace a complete pass of ascending bubble sort on [6,2,5,1].

Predict first

Compare (6,2), (6,5), then (6,1).

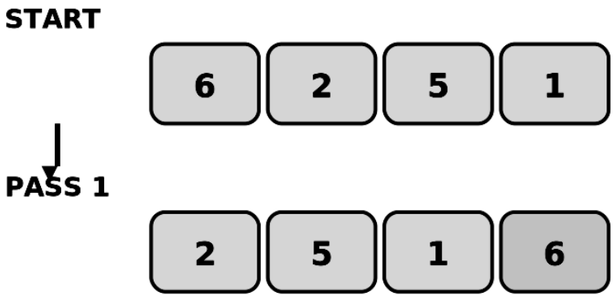
Model solution

After swaps: [2,6,5,1] → [2,5,6,1] → [2,5,1,6]. The last value is fixed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a complete pass of ascending bubble sort on [6,2,5,1].



Worked algorithm states: START 6,2,5,1; PASS 1 2,5,1,6

WE DO — GUIDED PRACTICE

Try → Discuss → Improve